

Incremental Spectral Sparsification for Large-Scale Graph-Based Semi-Supervised Learning

Daniele Calandriello · Alessandro Lazaric · Michal Valko · Ioannis Koutis

DALI 2016 poster — archival repo-owned render reconstructed from the original research source

Motivation & setting

- Graph-based semi-supervised learning exploits abundant unlabeled samples through a similarity graph, but exact harmonic-function methods scale poorly on large graphs.
- Goal: reduce memory and running time without compromising the predictive accuracy or the graph spectrum that learning depends on.
- Transductive setting: a graph $G=(X,E)$, a small random labeled set S , and a large unlabeled set T . Stable-HFS regularizes the graph Laplacian solution.

Core question

Can a streaming sparsifier preserve the learning guarantees while keeping only near-linear memory?

Sparse-HFS

- Process the edge stream in blocks and repeatedly resparsify the union of the previous sparsifier and the new block.
- Estimate effective resistances with an SDD solver; use them as importance-sampling probabilities.
- Keep the intermediate graph spectrally close to the full graph at every resparsification.
- Center the labels, then solve Stable-HFS on the final sparse graph.

Output

A sparse graph H and an SSL prediction whose generalization error is controlled relative to the full-graph solution.

Theoretical guarantees

Spectral approximation

With high probability, every eigenvalue of the sparsified Laplacian stays within a $(1\pm\epsilon)$ factor of the corresponding eigenvalue of the original graph.

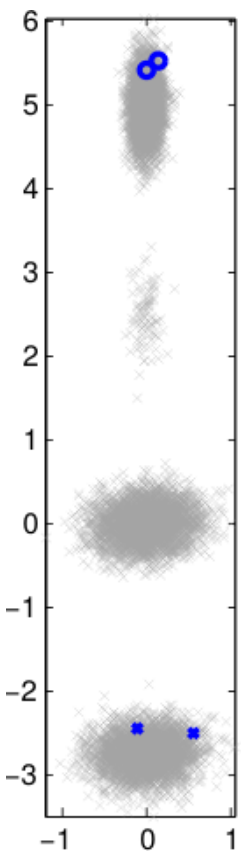
Learning guarantee

Because H is spectrally similar to G , the perturbation introduced in the Stable-HFS prediction can be bounded explicitly in the generalization error.

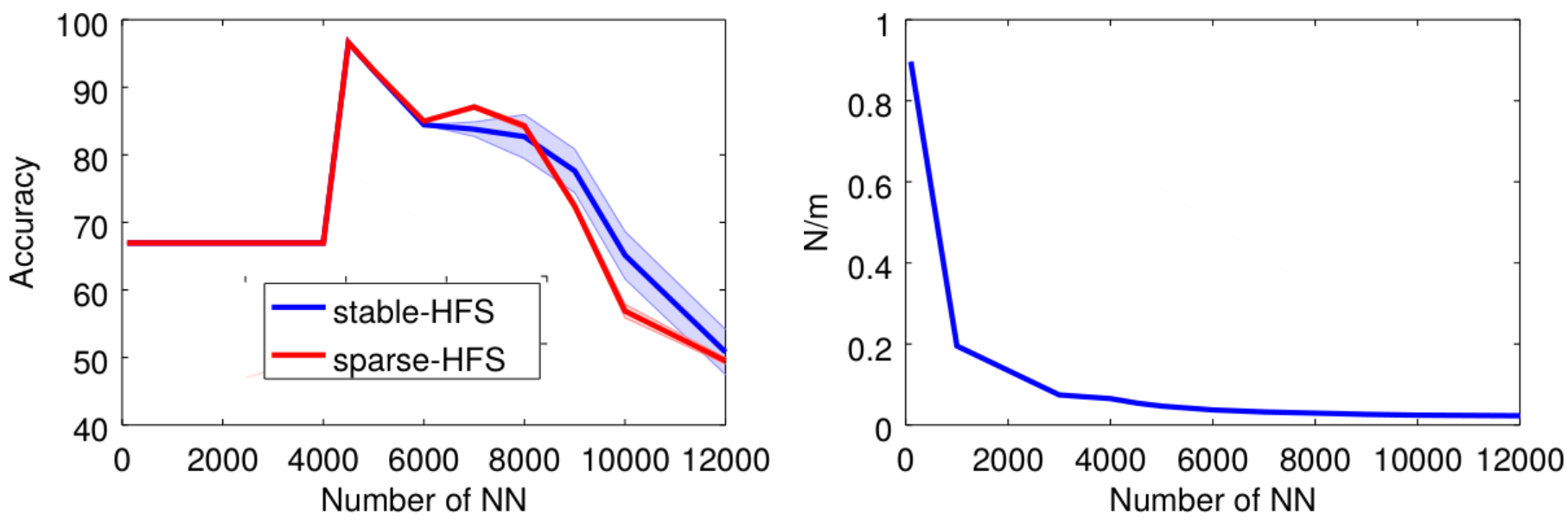
Complexity

- $O(n \log^2 n)$ memory for the sparsifier
- $O(m \log^3 n)$ preprocessing over the edge stream
- $O(n \log^3 n)$ final solve, up to the SDD-solver factors

Experiments



Toy graph: two positive and two negative clusters; only four labels.

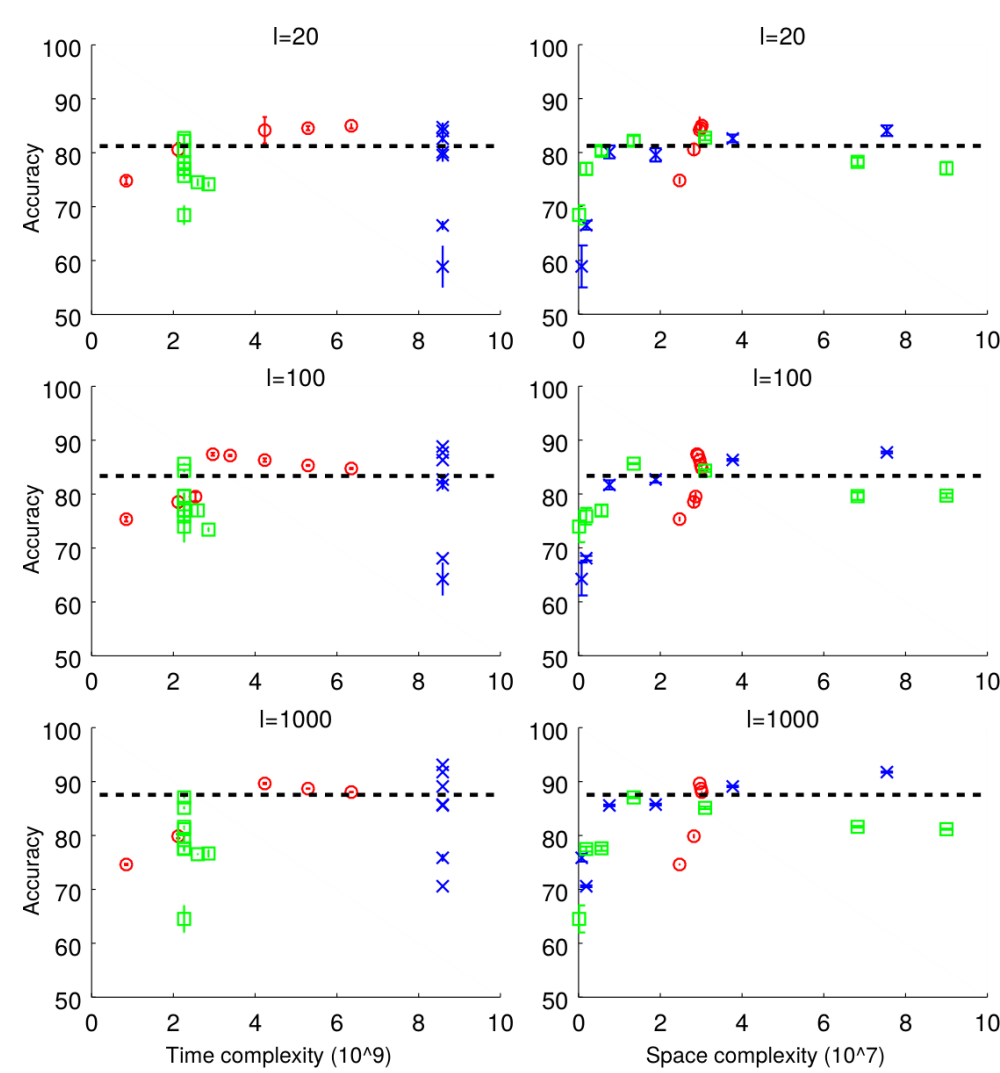


Sparse-HFS tracks Stable-HFS accuracy while the edge ratio shrinks as the graph densifies.

Why sparsification is the right abstraction

- Subsampling vertices can destroy the Laplacian geometry needed by graph-based learning.
- Spectral sparsification instead preserves all Laplacian quadratic forms simultaneously.
- This gives a direct bridge from graph approximation guarantees to learning guarantees.

Large-scale behavior



- TREC07 spam classification: sparse graph methods trade a small approximation error for major memory savings.
- As k grows, the full k -NN graph becomes dense while the sparsifier remains near-linear in n .

Streaming picture

edge blocks

$\Delta_1, \Delta_2, \dots$

resparsify

effective resistance

bounded H

spectral approximation

Stable-HFS

prediction

Key design constraints

- sparsifier size stays bounded
- intermediate approximation stays accurate
- amortized work remains polylogarithmic per edge

Complexity comparison

Method	Space	Preprocessing	Solve
Sparse-HFS	$n \log^2 n$	$m \log^3 n$	$n \log^3 n$
Stable-HFS	m	m	mn
Eigenfunctions	$nd+nq+b^2$	qb^3+db^3	q^3+nq
Subsampling	sk	m	s^2k+n

Takeaways

- Preserve the spectrum, not every edge.
- Streaming resparsification makes graph SSL practical at large scale.
- The approximation is accompanied by an explicit learning guarantee, not only a computational heuristic.
- The same spectral viewpoint later supports stronger ridge-aware graph learning methods.